

```
1 #include <stdio.h>
2
3 int main() {
4     int m, n, temp;
5
6     scanf("%d %d", &m, &n);
7
8     temp = m;
9     m = n;
10    n = temp;
11
12    printf("After Swapping: m value = %d; n value = %d", m, n);
13
14    return 0;
15 }
```

```
1 #include <stdio.h>
2
3 int main() {
4     int a, b;
5
6     scanf("%d %d", &a, &b);
7
8     printf("Sum = %d\n", a + b);
9     printf("Difference = %d\n", a - b);
10    printf("Product = %d\n", a * b);
11    printf("Quotient = %d\n", a / b);
12
13    return 0;
14 }
```

```
1 #include <stdio.h>
2
3 int main() {
4     float celsius, fahrenheit;
5
6     scanf("%f", &celsius);
7
8     fahrenheit = (celsius * 1.8) + 32;
9
10    printf("%.2f", fahrenheit);
11
12    return 0;
13 }
```

```
1 #include <stdio.h>
2
3 int main() {
4     float length, width, area;
5
6     scanf("%f %f", &length, &width);
7
8     area = length * width;
9
10    printf("The area of the rectangle is: %.1f", area);
11
12    return 0;
13 }
```

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5
6     scanf("%d", &n);
7
8     if (n % 2 == 0)
9         printf("Even");
10    else
11        printf("Odd");
12
13    return 0;
14 }
```

```
1 #include <stdio.h>
2
3 int main() {
4     int year;
5
6     scanf("%d", &year);
7
8     if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
9         printf("Leap Year");
10    else
11        printf("Not a Leap Year");
12
13    return 0;
14 }
```

```
1 #include <stdio.h>
2
3 int main() {
4     int a, b, c, max, min;
5
6     scanf("%d %d %d", &a, &b, &c);
7
8     max = a;
9     if (b > max) max = b;
10    if (c > max) max = c;
11
12    min = a;
13    if (b < min) min = b;
14    if (c < min) min = c;
15
16    printf("Maximum = %d\n", max);
17    printf("Minimum = %d", min);
```

QUESTIONS 8 / 8 completed

Sum-Avg

SUM-AVG

Write a program that takes a list of numbers as input and computes the sum and average of the numbers.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int a, b, c, max, min;
5
6     scanf("%d %d %d", &a, &b, &c);
7
8     max = a;
9     if (b > max) max = b;
10    if (c > max) max = c;
11
12    min = a;
13    if (b < min) min = b;
14    if (c < min) min = c;
15
16    printf("Maximum = %d\n", max);
17    printf("Minimum = %d", min);
18
```

INPUT

10 25 15

+ OR -

+ OR -

C program to Check
Whether a Number is
Positive or Negative.

Problem Description
The program takes the
given integer and
checks whether the
integer is positive
or negative.

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5
6     scanf("%d", &n);
7
8     if (n > 0)
9         printf("Positive");
10    else if (n < 0)
11        printf("Negative");
12    else
13        printf("Zero");
14
15    return 0;
16 }
```

65

Greatest

GREATEST

Write a program that takes two numbers as input and determines which one is larger.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int a, b;
5
6     scanf("%d %d", &a, &b);
7
8     if (a > b)
9         printf("%d", a);
10    else if (b > a)
11        printf("%d", b);
12    else
13        printf("Both are equal");
14
15    return 0;
16 }
```

Run

Save

10 20

Character ▾

CHARACTER

Write a program that takes a character as input and determines whether it is a vowel or a consonant.

PROGRAM EDITOR

Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     char ch;
5
6     scanf("%c", &ch);
7
8     if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' ||
9         ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U')
10         printf("Vowel");
11     else
12         printf("Consonant");
13
14     return 0;
15 }
```

b

Age

AGE

Write a program that takes an age as input and determines whether the person is a child, teenager, adult, or senior citizen.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int age;
5
6     scanf("%d", &age);
7
8     if (age < 13)
9         printf("Child");
10    else if (age < 20)
11        printf("Teenager");
12    else if (age < 60)
13        printf("Adult");
14    else
15        printf("Senior Citizen");
16
17    return 0;
18 }
```

Run **Save**

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PROGRAM EDITOR

40



Mohammad Asif K
192521122RD

UPS-LOWS

Write a program that takes a character as input and determines whether it is uppercase or lowercase.

```
1 #include <stdio.h>
2
3 int main() {
4     char ch;
5
6     scanf("%c", &ch);
7
8     if (ch >= 'A' && ch <= 'Z')
9         printf("Uppercase");
10    else if (ch >= 'a' && ch <= 'z')
11        printf("Lowercase");
12    else
13        printf("Invalid");
14
15    return 0;
16 }
```

A





SIMATS Online Programming Lab

C PROGRAMMING

Mohammad Asif K
192521122RD

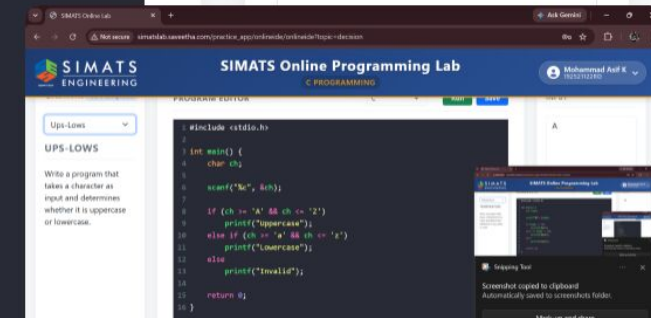
Multiple

MULTIPLE

Write a program that takes a number as input and determines whether it is a multiple of 3 or 5.

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5
6     scanf("%d", &n);
7
8     if (n % 3 == 0)
9         printf("Multiple of 3");
10    else if (n % 5 == 0)
11        printf("Multiple of 5");
12    else
13        printf("Not a multiple of 3 or 5");
14
15    return 0;
16 }
```

9

 Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

Grade

GRADE

Write a program that takes a grade as input and determines the corresponding letter grade (A, B, C, D, or F).

```
1 #include <stdio.h>
2
3 int main() {
4     int marks;
5
6     scanf("%d", &marks);
7
8     if (marks >= 90)
9         printf("A");
10    else if (marks >= 80)
11        printf("B");
12    else if (marks >= 70)
13        printf("C");
14    else if (marks >= 60)
15        printf("D");
16    else
17        printf("F");
18 }
```

95

 Snipping Tool

Screenshot copied to clipboard
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Mark-up and share

SUM(ODD,EVEN)

C Program to Find the Sum of Even and Odd Numbers.
Problem Description
The program takes the number N and finds the sum of odd and even numbers from 1 to N.

Testcases:

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5     int evenSum = 0, oddSum = 0;
6
7     scanf("%d", &n);
8
9     if (n < 1) {
10         printf("Invalid Input");
11         return 0;
12     }
13
14     for (i = 1; i <= n; i++) {
15         if (i % 2 == 0)
16             evenSum += i;
17         else
18             oddSum += i;
19     }
20
21     printf("Sum of Odd = %d\n", oddSum);
```

QUESTIONS 5 / 18 completed

Natural

NATURAL

Write a program that prints the first n natural numbers using a for loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5
6     scanf("%d", &n);
7
8     if (n <= 0) {
9         printf("Invalid Input");
10        return 0;
11    }
12
13    for (i = 1; i <= n; i++) {
14        printf("%d ", i);
15    }
16
17 }
```

INPUT

5

QUESTIONS 5 / 18 completed

Sum

SUM

Write a program that calculates the sum of the first n natural numbers using a while loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i = 1, sum = 0;
5
6     scanf("%d", &n);
7
8     if (n <= 0) {
9         printf("Invalid Input");
10        return 0;
11    }
12
13    while (i <= n) {
14        sum += i;
15        i++;
16    }
```

INPUT

5

QUESTIONS 5 / 18 completed

Table

TABLE

Write a program that prints the multiplication table of a given number using a for loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5
6     scanf("%d", &n);
7
8     if (n <= 0) {
9         printf("Invalid Input");
10        return 0;
11    }
12
13    for (i = 1; i <= 10; i++) {
14        printf("%d x %d = %d\n", n, i, n * i);
15    }
16 }
```

INPUT

5

Prime

PRIME

Write a program that takes a number as input and determines whether it is a prime number or not using a while loop.

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i = 2, prime = 1;
5
6     scanf("%d", &n);
7
8     if (n <= 1) {
9         printf("Not Prime");
10        return 0;
11    }
12
13    while (i <= n / 2) {
14        if (n % i == 0) {
15            prime = 0;
16            break;
17        }
18        i++;
19    }
20
```

7

Max

MAX

Write a C program to find the largest element in an array.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int arr[n];
8     int i;
9     int max;
10
11     for(i = 0; i < n; i++) {
12         scanf("%d", &arr[i]);
13     }
14
15     max = arr[0];
16     for(i = 1; i < n; i++) {
17         if(arr[i] > max) {
18             max = arr[i];
```

Run **Save**

INPUT

5
3 7 2 9 4

Average

AVERAGE

Write a C program to calculate the average of elements in an array.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int arr[n];
8     int i;
9     int sum = 0;
10    float avg;
11
12    for(i = 0; i < n; i++) {
13        scanf("%d", &arr[i]);
14        sum = sum + arr[i];
15    }
16
17    avg = (float)sum / n;
18
```

Run **Save**

INPUT

5
10 20 30 40 50

Reverse

REVERSE

Write a C program to reverse the elements of an array.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int arr[n];
8     int i;
9
10    for(i = 0; i < n; i++) {
11        scanf("%d", &arr[i]);
12    }
13
14    for(i = n - 1; i >= 0; i--) {
15        printf("%d ", arr[i]);
16    }
17
18    return 0;
```

Run Save

INPUT

5
1 2 3 4 5

2nd Max

2ND MAX

Write a C program to find the second largest element in an array.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int arr[n];
8     int i;
9     int max1, max2;
10
11     for(i = 0; i < n; i++) {
12         scanf("%d", &arr[i]);
13     }
14
15     max1 = max2 = arr[0];
16
17     for(i = 1; i < n; i++) {
18         if(arr[i] > max1) {
```

Run **Save**

INPUT

6
10 20 4 45 99 99

Sort A-Z

SORT A-Z

Write a C program to sort an array in ascending order.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int arr[n];
8     int i, j, temp;
9
10    for(i = 0; i < n; i++) {
11        scanf("%d", &arr[i]);
12    }
13
14    for(i = 0; i < n-1; i++) {
15        for(j = i+1; j < n; j++) {
16            if(arr[i] > arr[j]) {
17                temp = arr[i];
18                arr[i] = arr[j];
```

Kun **Save**

INPUT

5
5 2 8 1 9

Equal

EQUAL

Write a C program to check if two arrays are equal or not.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int arr1[n];
8     int arr2[n];
9     int i;
10    int equal = 1;
11
12    for(i = 0; i < n; i++) {
13        scanf("%d", &arr1[i]);
14    }
15
16    for(i = 0; i < n; i++) {
17        scanf("%d", &arr2[i]);
18    }
```

Run **Save**

INPUT

5
1 2 3 4 5
1 2 3 4 5

Frequency

FREQUENCY

Write a C program to find the frequency of an element in an array.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int arr[n];
8     int i;
9     int x;
10    int count = 0;
11
12    for(i = 0; i < n; i++) {
13        scanf("%d", &arr[i]);
14    }
15
16    scanf("%d", &x);
17
18    for(i = 0; i < n; i++) {
```

Run Save

INPUT

6
1 2 3 2 4 2
2

Duplicate

DUPLICATE

Write a C program to remove duplicate elements from an array.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int arr[n];
8     int temp[n];
9     int i, j;
10    int k = 0;
11
12    for(i = 0; i < n; i++) {
13        scanf("%d", &arr[i]);
14    }
15
16    for(i = 0; i < n; i++) {
17        int found = 0;
18        for(j = 0; j < k; j++) {
```

Run Save

INPUT

8
1 2 3 2 4 1 5 3

Insert

INSERT

Write a C program to insert an element at a specific position in an array.

PROGRAM EDITOR

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int arr[n+1];
8     int i;
9     int pos;
10    int element;
11
12    for(i = 0; i < n; i++) {
13        scanf("%d", &arr[i]);
14    }
15
16    scanf("%d %d", &pos, &element);
17
18    for(i = n; i >= pos; i--) {
```

Run Save

INPUT

5
10 20 30 40 50
3 25